



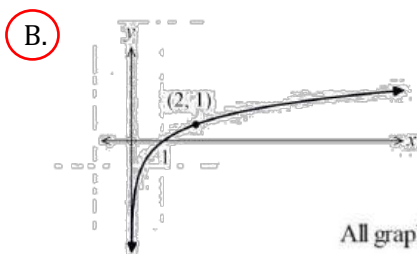
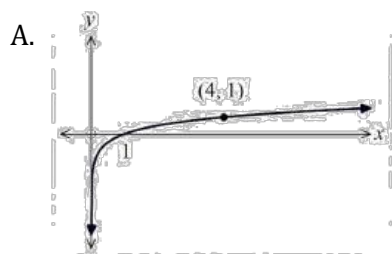
Exponential & Logarithmic Function

Topic Test 1 Solution

Total Marks: 30

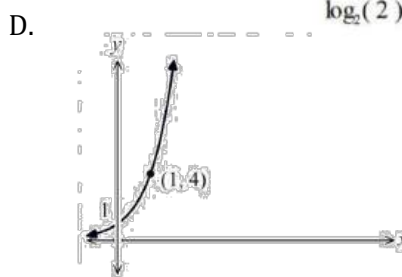
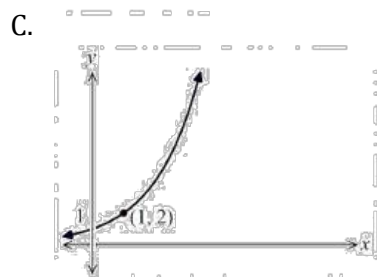
Time: 50 minutes

- 1 Solve the equation: $2\ln x = \ln(2x + 15)$. 2
- $$2\ln(x) = \ln(2x + 15)$$
- $$\ln(x^2) = \ln(2x + 15)$$
- $$x^2 = 2x + 15$$
- $$x^2 - 2x - 15 = 0$$
- $$(x - 5)(x + 3) = 0$$
- $$x = 5 \text{ or } x = -3$$
- But $\ln(-3)$ is not defined. $x = 5$ is the only solution
- 2 a. Evaluate $\log_9 3$. 1
- $$\log_9(3) = \log_9(\sqrt{9}) = \log_9\left(9^{\frac{1}{2}}\right) = \frac{1}{2}$$
- b. Evaluate $\log_5\left(\frac{1}{25}\right)$. 1
- $$\log_5\left(\frac{1}{25}\right) = \log_5\left(\frac{1}{5^2}\right) = \log_5(5^{-2}) = -2$$
- 3 Simplify $\frac{1}{2}\log_x 16 + 2\log_x 5$. 2
- $$\frac{1}{2}\log_x 16 + 2\log_x 5 = \log_x 16^{0.5} + \log_x 5^2$$
- $$= \log_x 4 + \log_x 25$$
- $$= \log_x 100 = 2\log_x 10$$
- 4 Find the x value when the graph of the function $f(x) = e^{2x} - 2$ intersects with the graph of $g(x) = e^x$. 2
- Solve equations simultaneously
- $$e^{2x} - 2 = e^x$$
- $$(e^x)^2 - e^x - 2 = 0$$
- $$(e^x - 2)(e^x + 1) = 0$$
- $$(e^x + 1) \neq 0$$
- Therefore
- $$(e^x - 2) = 0$$
- $$e^x = 2 \quad x = \log_e 2$$
- 5 What is the asymptote of $y = 4\log_2(5x) + 2$? 1
- $$5x > 0$$
- $$x > 0$$
- $\therefore$ Vertical asymptote is $x = 0$
- 6 Which sketch represents the curve $y = \log_2 x$? 1



All graphs $y = \log_a x$ have x intercept of 1

$\log_2(2) = 1$ so passes thru (2, 1)



- 7 The number of bats, N , in a park at time t years, is given by the formula: $N = N_0 e^{0.347t}$ where N_0 is a constant. Initially there were 40 bats in the park.

a. Find the value of N_0 .	(a)	$N = N_0 e^{0.347t}$ At $t = 0, N = 40$ $\therefore N_0 = 40$	1
	(b)	$N = 80$ $N = N_0 e^{0.347t}$ $80 = 40 \times e^{0.347t}$ $e^{0.347t} = 2$ $0.347t = \log_e 2$ $t = \frac{\log_e 2}{0.347} = 1.9975 \dots \approx 2$ years $\therefore$ It takes 2 years for the number of bats to reach 80.	2
c. How many bats were in the park after 12 years?	(c)	$t = 12$ $N = N_0 e^{0.347t} = 40 e^{0.347 \times 12} = 2573.132 \dots \approx 2573$ $\therefore$ There are 2573 bats	1
d. Find the rate of increase in the number of bats at 12 years.	(d)	Rate of increase is $\frac{dN}{dt}$ $\frac{dN}{dt} = 0.347 \times 40 e^{0.347t}$ $= 0.347 \times 40 e^{0.347 \times 12} = 892.877 \dots$ ≈ 893 bats per week	2
e. How many years would it take for the number of bats to increase by 600%? Answer correct to one decimal place.	(e)	600% increase = $6 \times 40 = 240$ or $N = 280$ $280 = 40 \times e^{0.347t}$ $e^{0.347t} = 7$ $0.347t = \log_e 7$ $t = \frac{\log_e 7}{0.347} = 5.6078 \approx 5.6$ years	1

- 8 Given that $f(x) = 3e^{4x}$. Find the exact value of $f'(-4)$. 2

$$\begin{aligned}
 f(x) &= 3e^{4x} \\
 f'(x) &= 4 \times 3e^{4x} = 12e^{4x} \\
 f'(-4) &= 12e^{4 \times -4} \\
 &= 12e^{-16} = \frac{12}{e^{16}}
 \end{aligned}$$

- 9 The number of bats, N , in a park at time t years, is given by the formula: $N = N_0 e^{0.347t}$ where N_0 is a constant. Initially there were 40 bats in the park.

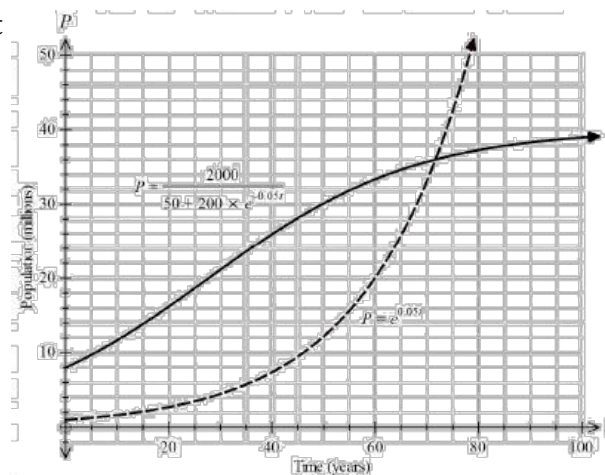
- a. How many bats were in the park after 12 years? 1

$$\begin{aligned}
 t &= 12 \\
 N &= N_0 e^{0.347t} \\
 &= 40 e^{0.347 \times 12} \\
 &= 2573.132 \dots \approx 2573 \\
 \therefore \text{There are 2573 bats}
 \end{aligned}$$

- b. How many years would it take for the number of bats to reach 80? Answer to the nearest year. $N = 80$ 2

$$\begin{aligned}
 N &= N_0 e^{0.347t} \\
 80 &= 40 \times e^{0.347t} \\
 e^{0.347t} &= 2 \\
 0.347t &= \log_e 2 \\
 t &= \frac{\log_e 2}{0.347} = 1.9975 \dots \approx 2 \text{ years} \\
 \therefore \text{It takes 2 years for the number of bats to reach 80.}
 \end{aligned}$$

- 10 The graph below shows two curves that represent models for the populations of two countries over a 100-year period. The equations of the two curves are given on the diagram.



- a. State which model has an upper limit to the population and give an estimate of that upper limit.

$$y = \frac{2000}{50 + 200e^{-0.05t}} \text{ has an upper limit}$$

as $t \rightarrow \infty$

$$e^{-0.05t} \rightarrow 0$$

so $200e^{-0.05t} \rightarrow 0$

$$\frac{2000}{50 + 200e^{-0.05t}} \rightarrow \frac{2000}{50 + 0} = \frac{200}{50} = 40$$

The upper limit is 40 million

- b. By solving the equations simultaneously, find the number of years (correct to 2 decimal places) after which the two models predict the population to be the same.

$$\frac{2000}{50 + 200e^{-0.05t}} = e^{0.05t}$$

$$2000 = e^{0.05t}(50 + 200e^{-0.05t})$$

$$2000 = (50e^{0.05t} + 200 \times e^{0.05t} \times e^{-0.05t})$$

$$2000 = 50e^{0.05t} + 200$$

$$50e^{0.05t} = 1800$$

$$e^{0.05t} = 36$$

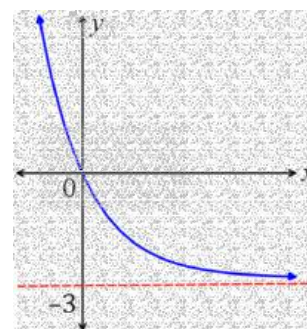
take $\ln$ of both sides

$$0.05t = \ln(36)$$

$$t = \frac{\ln(36)}{0.05} = 20\ln(36) = 71.67 \text{ years (2 dp)}$$

- 11 The graph of the function $y = ae^{-x} + b$ is shown below. What are the values of a and b ?

- A. $a = -3$ and $b = -3$ B. $a = -3$ and $b = 3$
C. $a = -3$ and $b = 0$ D. $a = 3$ and $b = -3$



The related graph $y = ae^{-x}$ is translated 3 units below the x-axis.

$$\therefore y = ae^{-x} - 3$$

The graph passes through the point (0, 0)

$$y = ae^{-x} - 3$$

$$0 = a \times e^0 - 3 \text{ hence } a = 3$$

$$\therefore a = 3 \text{ and } b = -3$$

- 12 Simplify $\log(a^2) + \log(b^2) - 2\log(ab)$.

$$\log(a^2) + \log(b^2) - 2\log(ab) = \log \frac{a^2 \times b^2}{(ab)^2} = \log 1 = 0$$

13 Solve $\log_2(x+1) - \log_2(x-1) = 2$.

2

$$\log_2\left(\frac{x+1}{x-1}\right) = 2 \quad \text{for } x+1 > 0 \text{ and } x-1 > 0$$

$$\therefore \frac{x+1}{x-1} = 4, \quad x > 1$$

$$\therefore x > 1 \text{ and } x+1 = 4x-4$$

$$5 = 3x$$

$$\therefore x = \frac{5}{3}$$

14 Solve $4^{x-2} = 2^{3x-4}$.

$$4^{x-2} = 2^{3x-4}$$

$$(2^2)^{x-2} = 2^{3x-4}$$

$$2^{2x-4} = 2^{3x-4}$$

$$\therefore 2x-4 = 3x-4$$

$$x = 0$$

1



Exponential & Logarithmic Function

Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 Which of the following functions is NOT even?

A. $f(x) = x^{20}$

B. $f(x) = 3|x|$

C. $f(x) = \log_2 x$

D. $f(x) = 2^x + 2^{-x}$

Check $f(-x) \neq f(x)$

$(-x)^{20} = x^{20}$, even

$2^{-x} + 2^x = 2^x + 2^{-x}$, even

$\log_2(-x) \neq \log_2 x$, not even

$3|-x| = 3|x|$, even

1

2 Given that $\log_7 3 = a$ and $\log_7 8 = b$, express the following in terms of a and b .

a. $\log_7 63$

(a) $\log_7 63 = \log_7 (9 \times 7)$
 $= \log_7 3^2 + \log_7 7$
 $= 2a + 1$

2

b. $\log_7 \frac{16}{27}$

(b) $= \log_7 16 - \log_7 27$
 $= \log_7 8^{\frac{2}{3}} - \log_7 3^3$
 $= \frac{2}{3} \log_7 8 - 3 \log_7 3$
 $= \frac{2}{3} b - 3a$

2

3 Which of the following is equivalent to $\ln(e^{x^2})$?

A. x^2

B. x

C. $2x$

D. 2

$\ln e^{x^2} = x^2 \times \ln e = x^2 \times 1 = x^2$ (A)

1

4 a. By writing 3^x as $e^{f(x)}$ where $f(x)$ is a particular function, differentiate 3^x .

$\frac{d}{dx} [3^x]$
 $= \frac{d}{dx} [e^{\ln 3^x}]$
 $= \frac{d}{dx} [e^{x \ln 3}]$
 $= \ln 3 \times e^{x \ln 3}$
 $= \ln 3 \times e^{\ln 3^x}$
 $= (\ln 3) 3^x$

2

b. Hence, find a formula for $\frac{d}{dx}(a^x)$ where a is a positive number.

$\frac{d}{dx} [a^x]$
 $= \frac{d}{dx} [e^{\ln a^x}]$
 $= \frac{d}{dx} [e^{x \ln a}]$
 $= (\ln a) \times e^{x \ln a}$
 $= (\ln a) \times a^x$

2

5 a. Simplify $\ln(a-b) + \ln(a+b)$ for positive values of a and b where $a > b$.

$\ln(a-b) + \ln(a+b) = \ln[(a-b)(a+b)] = \ln(a^2 - b^2)$

2

b. Hence, or otherwise, simplify $\ln(a-b) + \ln(a+b) + \ln(a^2 + b^2)$.

$\ln(a-b) + \ln(a+b) + \ln(a^2 + b^2)$
 $= \ln[(a-b)(a+b)(a^2 + b^2)]$
 $= \ln(a^4 - b^4)$

2

6 It is given that $\ln(\ln a) = \ln(\ln b) + \ln(\ln c) + \ln(\ln d)$. Which statements is true? 1

- A. $a = b + c - d$ B. $a = \frac{bc}{d}$ $\ln \ln a = \ln \ln b + \ln \ln c - \ln \ln d$
 $\ln \ln a = \ln \ln \frac{bc}{d}$ (log laws)
 C. $\ln(\ln a) = \frac{\ln(\ln b) \times \ln(\ln c)}{\ln(\ln d)}$ D. $\ln(\ln a) = \frac{b+c}{d}$ $a = \frac{bc}{d}$

7 On the Richter scale, the magnitude R of an earthquake of intensity I is given by the formula

$$R = \log_{10} \left(\frac{I}{I_0} \right), \text{ where } I_0 \text{ is a reference intensity.}$$

- a. An earthquake of magnitude 6.0 can damage well built structures. Express its intensity in terms I_0 . 2
 $6 = \log_{10} \left(\frac{I}{I_0} \right) \Rightarrow \frac{I}{I_0} = 10^6 \Rightarrow I = 10^6 I_0 \text{ or } 1000000 I_0$
 b. Consider an earthquake that is five times more intense than the one in part (a). Find its magnitude correct to one decimal place. 2
 $I = 5 \times 10^6 I_0 \Rightarrow R = \log_{10} \left(\frac{5 \times 10^6 I_0}{I_0} \right)$
 $= \log_{10}(5 \times 10^6) = 6.7 \text{ (1 dp)}$

8 A cup of tea is left on a table. The temperature of the tea T ($^{\circ}\text{C}$) after t minutes is given by the formula $T = 90e^{-0.006t}$

- a. Use differentiation to explain why the temperature of the tea is decreasing. 2

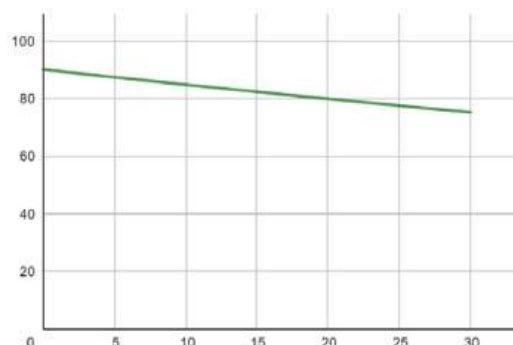
$$\frac{dT}{dt} = -0.54e^{-0.006t} < 0 \text{ for all } t$$

Therefore the temperature is decreasing

- b. What is the initial temperature of the tea? $T(0) = 90$ 1
 c. What will be the temperature of the tea be after 5 minutes? Give your answer correct to the nearest $^{\circ}\text{C}$. $T(5) = 87 \text{ (0 dp)}$ 1
 d. How long does it take for the temperature of the tea to be 45°C ? Give your answer correct to the nearest minute. 1

$$T = 45 \Rightarrow e^{-0.006t} = 0.5 \Rightarrow t = -\frac{1}{0.006} \ln 0.5 = 116 \text{ (0 dp)}$$

- e. Sketch a graph of the temperature of the tea for $0 \leq t \leq 30$ in the space below. 1



- f. Explain if the model is realistic. 1

The model is not realistic since it will take too much time for the cup of tea to cool completely.

- 9 A certain radioactive substance decays such that its mass, M grams, after t hours is given by $M = 75e^{-kt}$, where k is a constant. A block of this substance had an original mass of 75 grams. After 10 hours, its mass had reduced to 62 grams.

a. Find the exact value of k .

$$\begin{aligned} \text{When } t=10, M=62 \\ 62 &= 75e^{-10k} \\ \frac{62}{75} &= e^{-10k} \\ \therefore k &= -\frac{1}{10} \ln\left(\frac{62}{75}\right) \end{aligned}$$

2

b. How much has the substance's mass reduced after 5 hours? Give your answer correct to one decimal place.

$$\begin{aligned} \text{When } t=5, M &= 75e^{\frac{1}{10} \ln\left(\frac{62}{75}\right) \times 5} = 68.2 \text{ g} \\ \therefore \text{Mass reduced} &= 75 - 68.2 = \boxed{6.8 \text{ g}} \end{aligned}$$

2